MIL-C-18438B(SHIPS)
AMENDMENT - 1
26 November 1962

MILITARY SPECIFICATION

CHARGERS, BATTERY, GENERATOR TYPE

This amendment forms a part of Military Specification MIL-C-18438B (SHIPS), 14 June 1962.

Pages 1 and 2, paragraph 2.1: Under "SPECIFICATIONS, MILITARY" add:

"MIL-S-19500/124 - Semiconductor Diode, Silicon, 10-Watt, Voltage Regulator Types 1N2970B thru 1N3015B and 1N2970RB thru 1N3015RB."

Pages 1 and 2, paragraph 2.1: Under "STANDARDS, MILITARY", add:

"MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment."

Page 5, table II: In column 3, opposite "Coils (Single Layer), (b) Class B insulation, Method 2", delete "70" and substitute "90".

Page 6, paragraph 3.2.16, last sentence: Delete "200 r.p.m." and substitute "140 percent".

Pages 8 and 9, paragraphs 3.2.18 through 3.2.21: Delete and substitute:

"3.2.18 Electron tubes and semiconductor devices.-

"3.2.18.1 The use of electron tubes will not be permitted.

"3.2.18.2 Use of controlled rectifiers is permitted only when specifically approved on an individual application basis by the bureau or agency concerned. Approval of controlled rectifiers shall be obtained prior to considering them for use in a design.

"3.2.18.3 Prior to ordering semiconductor devices, an Electron Tube and Semiconductor Device Complement Report, as required by MIL-E-917, shall be submitted for bureau or agency approval.

"3.2.18.4 Rectifiers.-

"3.2.18.4.1 Rectifiers shall be either silicon or selenium types. When both types are suited to an application, the preference shall be silicon over selenium.

FSC 6130

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"3.2.18.4.2 Selenium rectifiers.- Selenium rectifiers shall be supplied in accordance with the requirements of MIL-R-18281.

"3.2.18.4.3 Silicon rectifiers.- Individual silicon rectifiers shall be selected from MIL-STD-701. In addition, they shall comply with the requirements of the individual detail specifications and maximum rating and rerating values specified in MIL-STD-701. Non-listed silicon rectifiers shall require specific approval as required by Specification MIL-E-917 for nonstandard parts. In addition they shall either be covered by an individual detail specification, or they shall comply with the requirements of an individual detail specification listed in MIL-STD-701 for a similarly constructed device (axial lead, stud-mounted, glass package, and so forth) except that the device manufacturer's latest published commercial specification values applicable to the device (or approved values based on the commercial product) shall be substituted for the characteristic value contained in the applicable specification sheet. Diodes shall be individually replaceable.

"3.2.18.5 Voltage reference diodes.- Voltage reference diodes shall be selected from MIL-STD-701. They shall comply with the requirements specified in applicable MIL-S-19500 detail specifications, Nominal 10-watt, 5-to 6-volt regulator diodes having substantially zero temperature coefficient are also permitted provided they are procured to MIL-S-19500/124, except that the device manufacturer's latest published commercial specification values applicable to the device (or approved values based on the commercial product) shall be substituted for the characteristic values contained in that specification; provided also, that special selection for temperature coefficient of voltage is not required. All other devices shall require specific approval as required by MIL-E-917 for nonstandard parts; in addition, they shall either be covered by an individual specification sheet or they shall comply with the requirements of an individual detail specifications listed in MIL-STD-701 for a similarly constructed device substituting characteristic values peculiar to the devices to be used. Diodes shall be individually replaceable.

"3.2.18.6 Voltage regulator devices.- Voltage regulator devices shall be selected from MIL-STD-701 and supplied in accordance with the requirements of MIL-S-19500 and applicable detail specifications thereunder as specified in MIL-STD-701. The temperature coefficient of voltage is not a controlled characteristic; device usage which requires control of this characteristic shall require the device to be treated as a nonstandard part. Non-listed devices shall also be treated as nonstandard parts, and shall require specific approval as required by MIL-E-917. In addition, non-listed devices shall either be covered by an individual detail specification, or they shall comply with the requirements of an individual detail specification listed in MIL-STD-701 for a similarly constructed device, except that the device manufacturer's latest published commercial specification values applicable to the device (or approved values based on the commercial product) shall be substituted for the characteristic values contained in the applicable specification. Voltage regulator diodes shall be individually replaceable.

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"3.2.18.7 Transistors.- Individual transistors shall be selected from the silicon types in MIL-STD-701. All transistors shall be procured to the requirements of MIL-S-19500 and, where issued, to individual Military detail specifications under MIL-S-19500. If not specifically covered by such a detail specification, they shall comply with the inspection procedures of a detail specification listed in MIL-STD-701 for a similarly constructed device, except that the manufacturer's latest commercial specification values applicable to the device (or approved values based on the commercial product) shall be substituted for the characteristic values contained in the applicable detail specifications. All other types will require specific approval as required by MIL-E-917 for nonstandard parts. The requirements specified for preferred types shall apply for all transistors approved for use in this equipment. Transistors shall be individually replaceable."

Page 10, paragraph 3.2.24.5: Add "(as specified, see 6.1)."

Page 11: Add as paragraph 3.3.3.1.1(d)(10)h:

"(h) Location of the center of gravity in two views."

Page 18, paragraph 4.5.4: Delete all information after the first sentence and substitute "Acceptability of totally enclosed machines shall be determined by tests for spraytight enclosures as specified in MIL-STD-108."

Page 19, paragraph 4.5.12, line 7: Delete "herein" and substitute "in 4.5.11".

Page 20, paragraph 4.5.18, first sentence: Delete "3.2.17.13" and substitute "3.2.17.14."

Page 21, paragraph 6.1: Add as items (p) and (q):

"(p) Where grounded system is desired, specify whether negative or positive ground is required (see 3.2.4).
"(q) Specify whether voltage regulator and current limiting elements' design should be static, vibrating contact, or carbon pile (see 3.2.24.5)."

Pages 23 through 27, Appendix I: Delete.

Preparing activity:
Navy - Ships
(Project 6130-N048Sh)

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